

Assignment -1
Department of Computer Science and Engineering (CSE)
3rd Semester
Subject: Theory of Computation (CSE 0613-2101)

Deadline: 24/02/26

[Answer all the questions on A4 size paper with sufficient margins on all sides and proper cover page. ****Note:** Penalties apply for late submission.]

1. a) Define Finite Automata.
b) Draw a Deterministic Finite Automaton (DFA) that accepts all strings over $\{0,1\}$ that end with 01.
2. Write a Regular Expression for the language over $\{a, b\}$ that contains all strings starting with a and ending with b.
3. a) What is the difference between a DFA and an NFA?
b) Design an NFA with $\Sigma = \{0, 1\}$ in which double '1' is followed by double '0'.
4. Convert the following Regular Expression into an equivalent NFA:
$$(0 + 1)^*01$$
5. Convert the given NFA with states $\{A, B, C, D\}$, input alphabet $\{0,1\}$, start state A, and accepting state D into an equivalent DFA using the subset construction method. Clearly show the DFA states (as subsets of NFA states), the transition table, and identify the accepting states.

(**Note: follow this url for conversion process guide: <https://www.geeksforgeeks.org/theory-of-computation/conversion-from-nfa-to-dfa/>)

